

Week 5: Real-world data integration

Data sources

We selected two complementary openFDA datasets: 100 methylphenidate product listings from the NDC Directory and 66 methylphenidate recall records. Product listings describe medicines, while recall records describe events; neither dataset is a subset of the other.

The product dataset was updated on 7 October 2026 and the recall dataset on 30 September 2026. Both were accessed on 7 October 2026, require no payment or registration, and use [openFDA's CC0 licence](#). The product selection contains 100 of 262 matching records; the recall selection contains all 66 matches.

Sources: [product data](#) and [recall data](#). Selected fields are included in the import SQL.

Cleaning and schema

We added two tables: `FDA_Product`, identified by product NDC, and `FDA_Recall`, identified by recall number. The original prescription tables remain unchanged and fictional. We did not link FDA records to mock prescriptions because no verified mapping exists.

No missing selected values or duplicate identifiers were found. Dates were converted from YYYYMMDD to SQL dates. Original names and punctuation were preserved, so company-name variants remain separate labels. Required fields, primary keys and a recall-classification check enforce valid records. Repeating the import refreshes only the FDA tables.

Normalization

Both tables meet 1NF because their fields contain single values; nested arrays are excluded. Each has a single-column primary key, with no assumed partial dependencies, satisfying 2NF. No further dependencies between non-key fields were identified, so the selected design meets 3NF under these assumptions. No additional decomposition was needed.

Queries and validation

Testing on MySQL 8.4.0 confirmed that all 166 imported records matched the selected source values. Repeated imports preserved counts, while duplicate product keys and invalid recall classifications were rejected.

The Week 3 queries still returned the expected fictional results. Four adapted queries returned 100 product listings, seven dosage-form groups, 53 Class II and 13 Class III recalls, and 14 latest recalls grouped by exact company label. These results matched expectations.

Limitations and future work

The data concerns US products and does not measure student misuse. Product listing does not establish FDA approval, and recall counts do not measure prescribing frequency. We still lack motives, actual consumption, diagnoses and outcomes.

The Week 4 goals of testing more examples and checking accuracy were addressed. Comparing changes over time and collecting student and staff feedback remain next steps. Longer-term plans include voluntary surveys, motives and outcomes, and protected student data.

Reproduction

After the Week 3 setup, run SQL file 05, rerun files 03 and 04, then run file 06. The import contains the selected data, so no live API connection is needed.